EXPERIMENT 9: Install a Hadoop Single-Node Cluster on Windows and Execute Word Count

1. Aim

To install and configure a single-node Hadoop environment on Windows and execute the Word Count MapReduce application.

2. Procedure

1. Install a compatible Java Development Kit on Windows.
2. Set JAVA_HOME to the JDK installation directory.
3. Download a Hadoop binary distribution and extract it, for example, to C:\\hadoop.
4. Set HADOOP_HOME and add the Hadoop bin directory to the system PATH.
5. Configure core-site.xml, hdfs-site.xml, mapred-site.xml and yarn-site.xml.
6. Configure Windows-compatible Hadoop native binaries, including a version-compatible winutils.exe where required.
7. Create the required NameNode and DataNode storage directories.
8. Format the NameNode from the Hadoop command prompt.
9. Start HDFS and YARN using the Windows command scripts supplied by the distribution.
10. Create an HDFS input directory using hdfs dfs -mkdir -p /input.
11. Create a text file named input.txt and enter sample words.
12. Upload the file to HDFS using hdfs dfs -put input.txt /input.
13. Run the Hadoop MapReduce examples JAR with the wordcount command.
14. Display the output using hdfs dfs -cat /output/part-r-00000.

3. Program

Sample input file – input.txt:

```
hello world
hello hadoop
hadoop is useful
```

Hadoop Word Count command (adjust the JAR version/path as installed):

```
hdfs dfs -mkdir -p /input
hdfs dfs -put input.txt /input

hadoop jar %HADOOP_HOME%\share\hadoop\mapreduce\hadoop-mapreduce-examples-3.3.6.jar wordcount /input /output

hdfs dfs -cat /output/part-r-00000
```

4. Output of Program

```
hadoop  2
hello   2
is      1
useful  1
world   1
```

Note: Windows Hadoop setup requires native binaries compatible with the exact Hadoop version. The example JAR path and version must match the installed distribution.

5. Result

Thus, Hadoop was configured in single-node mode on Windows, and the Word Count MapReduce application was executed successfully.

EXPERIMENT 9: Install a Hadoop Single-Node Cluster on Windows and Execute Word Count

1. Aim

To install and configure Apache Hadoop in a single-node environment on Microsoft Windows and execute the Word Count MapReduce application to count the frequency of words in an input text file.

2. Procedure

1. Start the Windows operating system and open a web browser. 2. Download and install a compatible Java Development Kit. 3. Configure the JAVA_HOME environment variable to point to the JDK installation directory. 4. Verify Java installation by executing java -version in the command prompt. 5. Download the Apache Hadoop binary distribution and extract it into a suitable directory, such as C:\hadoop. 6. Configure the HADOOP_HOME environment variable to point to the Hadoop installation directory. 7. Add the Hadoop bin directory to the system PATH. 8. Open the Hadoop configuration directory and edit the required configuration files. 9. Configure core-site.xml to specify the default filesystem URI. 10. Configure hdfs-site.xml to set the replication factor to 1 for the single-node cluster. 11. Configure mapred-site.xml to use YARN as the MapReduce execution framework. 12. Configure yarn-site.xml to enable the MapReduce shuffle service. 13. Configure the Windows-compatible Hadoop native binaries, including a version-compatible winutils.exe where required. 14. Create the required NameNode and DataNode storage directories. 15. Open the Hadoop command prompt and format the NameNode to initialize the HDFS filesystem. 16. Start the HDFS and YARN services using the Windows scripts provided by the Hadoop distribution. 17. Verify the running services and ensure that the cluster is ready to accept jobs. 18. Create a sample text file named input.txt containing a few lines of text. 19. Create an HDFS directory named /input using the hdfs dfs -mkdir command. 20. Upload the input file to HDFS using hdfs dfs -put input.txt /input. 21. Execute the Hadoop MapReduce Word Count example using the appropriate examples JAR file. 22. Specify /input as the input directory and /output as the output directory. 23. After successful execution, retrieve the result using hdfs dfs -cat /output/part-r-00000. 24. Observe the word-frequency output and verify that the number of occurrences is correct.

4. Output of Program

Input File – input.txt: hello world hello hadoop hadoop is useful Expected Output: hadoop 2 hello 2 is 1 useful 1 world 1 Explanation: The Word Count application uses the MapReduce programming model. - Map Phase: Reads the input text and generates key-value pairs in which each word is associated with the value 1. - Reduce Phase: Groups identical words and adds their corresponding values to calculate the total frequency. For example, the word hello appears twice, so its final count is 2.

5. Result

Thus, Apache Hadoop was installed and configured in single-node mode on Windows. The input text file was uploaded to HDFS, and the Word Count MapReduce application was executed successfully to calculate the frequency of each word.